Day 3: Conversion of Borneol to Camphor

Ellie Campbell

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Introduction

In this experiment, borneol was converted to camphor. Reflux was utilized to initiate the reaction and the final product was isolated using a water bath.

Materials and Methods

First, 0.3g of the borneol product from day 2 of the experiment was added to a long neck round bottom flask with 0.3 mL of glacial acetic acid. The borneol was then dissolved in 1 mL of acetone and a boiling chip was added to the mixture. A reflux was then assembled using a distillation column and the apparatus was placed in a water bath. The water bath temperature was kept between 40 and 50 degrees Celsius. After all the reagents dissolved, 6 mL of aqueous sodium hypochloride solution was added dropwise for 20 minutes. After each addition of hypochloride, the mixture was gently swirled, and the reaction was allowed to continue for 15 minutes after the last addition of hypochloride. Next, the mixture was allowed to cool to room temperature and the camphor was extracted using two 2 mL portions of methylene chloride. After the extraction, the solution was neutralized using saturated sodium bicarbonate. Sodium bicarbonate was added until no bubbles were produced after each addition. The organic solution was then washed with 2 mL of 5% sodium bisulfate and 2 mL of water. The organic solution was then dried using anhydrous sodium sulfate. The methylene chloride was then evaporated over a water bath and the collected crystals were weighed. All remaining glassware was cleaned with soap and water and left to air dry. The remaining waste was disposed of in appropriate waste containers.

Results

Weight of Watch Glass (g)	**35.791**
Weight of Watch Glass with Product (g)	**35.911**
Weight of Product (g)	**0.12**

Table 1: Weight of watch glass, weight of watch glass and product, weight of product

Calculations

Theoretical yield: 0.3g borneol/154.25g = 0.001945 mol Borneol = 0.001945 mol camphor

Actual yield: 0.12g camphor/152.23g = 0.000788 mol camphor

Percent yield: 0.000788/0.001945 * 100% = 40.5%

Discussion

In this experiment, borneol reacted with glacial acetic acid to produce camphor. Reflux was used to initiate the reaction. After reflux was complete, sodium hypochloride bleach was added to the reaction and used as the oxidizing agent. After the reaction with the bleach was complete, the camphor was extracted using DCM, and two layers were observed. Next, the organic layer was removed, and sodium bicarbonate was used to neutralize the methylene chloride. After the product was neutralized and washed, the DCM was evaporated over a water bath. The organic product, camphor, was isolated and formed crystals. The percent yield calculated for this experiment is 40.5%. Many inaccuracies could have contributed to the lower percent yield. For example, if the bleach was not mixed thoroughly and given enough time to react with the borneol, this could contribute to a lower yield. This could be prevented by swirling the mixture after each addition of bleach and allowing the reaction to continue for 15 minutes after the last addition of bleach. Additionally, leaving some of the organic product behind with the inorganic

product during extraction would also lead to a lower percent yield. This could be prevented by using a separatory funnel during extraction.

References

Keller L, Joseph K, and Miguel S. 2011. CHEM 2081 Organic Chemistry II Lab Manual [lab manual]. Auburn (Al): Auburn University.